*4.4 | THE HALL EFFECT

The Hall effect is a consequence of the forces that are exerted on moving charges by electric and magnetic fields. The Hall effect is used to distinguish whether a semiconductor is n type or p type[1] and to measure the majority carrier concentration and majority carrier mobility. The Hall effect device, as discussed in this section, is used to experimentally measure semiconductor parameters. However, it is also used extensively in engineering applications as a magnetic probe and in other circuit applications.

---

[1]We will assume an extrinsic semiconductor material in which the majority carrier concentration is much larger than the minority carrier concentration.

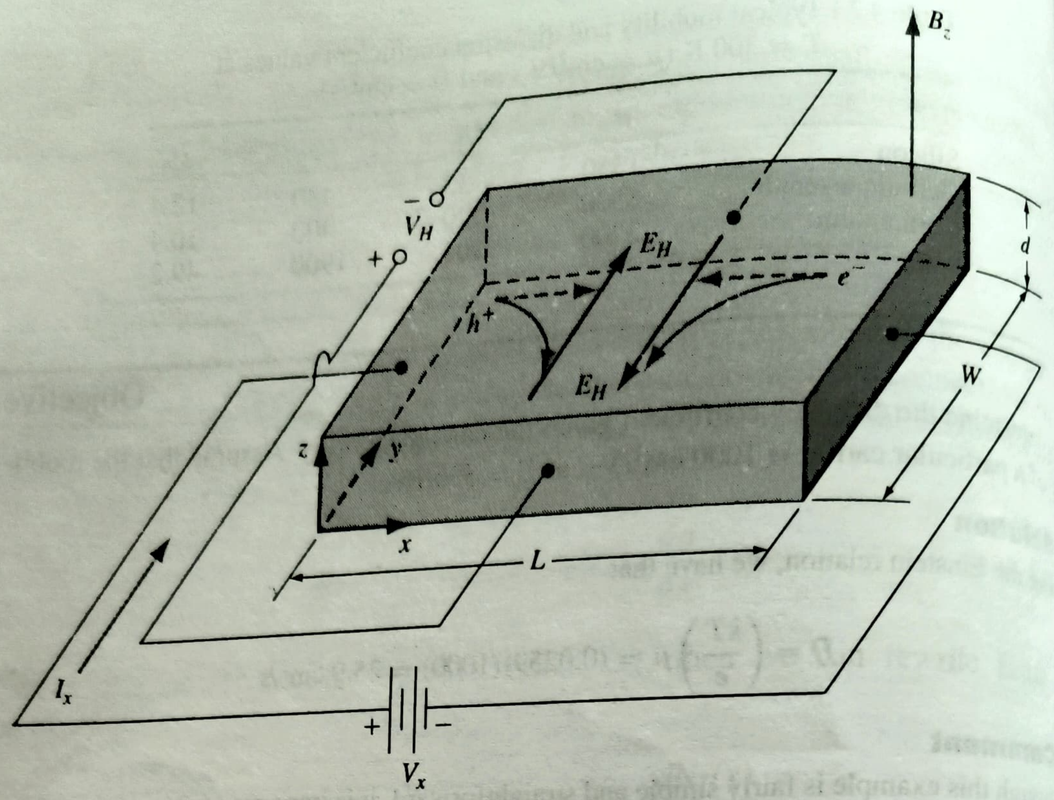

Figure 4.13 | Geometry for measuring the Hall effect.

The force on a particle having a charge q and moving in a magnetic field is given by

$$F = qv \times B \tag{4.46}$$

where the cross product is taken between velocity and magnetic field so that the force vector is perpendicular to both the velocity and magnetic field.

Figure 4.13 illustrates the Hall effect. A semiconductor with a current I_x is placed in a magnetic field perpendicular to the current. In this case, the magnetic field is in the z direction. Electrons and holes flowing in the semiconductor will experience a force as indicated in the figure. The force on both electrons and holes is in the $(-y)$ direction. In a p-type semiconductor ($p_0 > n_0$), there will be a buildup of positive charge on the $y = 0$ surface of the semiconductor and, in an n-type semiconductor ($n_0 > p_0$), there will be a buildup of negative charge on the $y = 0$ surface. This net charge induces an electric field in the y-direction as shown in the figure. In steady state, the magnetic field force will be exactly balanced by the induced electric field force. This balance may be written as

$$F = q[\mathrm{E} + v \times B] = 0 \tag{4.47a}$$

which becomes

$$q\mathrm{E}_y = qv_x B_z \tag{4.47b}$$

The induced electric field in the y-direction is called the *Hall field*. The Hall field produces a voltage across the semiconductor which is called the *Hall voltage*. We can write

$$V_H = +\mathrm{E}_H W \tag{4.48}$$

where E_H is assumed positive in the $+y$ direction and V_H is positive with the polarity shown.

In a p-type semiconductor, in which holes are the majority carrier, the Hall voltage will be positive as defined in Figure 4.13. In an n-type semiconductor, in which electrons are the majority carrier, the Hall voltage will have the opposite polarity. The polarity of the Hall voltage is used to determine whether an extrinsic semiconductor is n-type or p-type.

Substituting Equation (4.48) into Equation (4.47) gives

$$V_H = \upsilon_x W B_z \tag{4.49}$$

For a p-type semiconductor, the drift velocity of holes can be written as

$$\upsilon_{dx} = \frac{J_x}{ep} = \frac{I_x}{(ep)(Wd)} \tag{4.50}$$

where e is the magnitude of the electronic charge. Combining Equations (4.50) and (4.49), we have

$$V_H = \frac{I_x B_z}{epd} \tag{4.51}$$

or, solving for the hole concentration, we obtain

$$p = \frac{I_x B_z}{ed V_H} \tag{4.52}$$

The majority carrier hole concentration is determined from the current, magnetic field, and Hall voltage.

For an n-type semiconductor, the Hall voltage is given by

$$V_H = -\frac{I_x B_z}{ned} \tag{4.53}$$

so that the electron concentration is

$$n = -\frac{I_x B_z}{ed V_H} \tag{4.54}$$

Note that the Hall voltage is negative for the n-type semiconductor; therefore, the electron concentration determined from Equation (4.54) is actually a positive quantity.

Once the majority carrier concentration has been determined, we can calculate the low-field majority carrier mobility. For a p-type semiconductor, we can write

$$J_x = ep\mu_p \mathrm{E}_x \tag{4.55}$$

The current density and electric field can be converted to current and voltage so that Equation (4.55) becomes

$$\frac{I_x}{Wd} = \frac{ep\mu_p V_x}{L} \tag{4.56}$$

The hole mobility is then given by

$$\mu_p = \frac{I_x L}{ep V_x W d} \tag{4.57}$$

Similarly for an n-type semiconductor, the low-field electron mobility is determined from

$$\mu_n = \frac{I_x L}{en V_x W d} \tag{4.58}$$

in s, and m^* is in kg, so that μ comes out in $m^2/(V.s)$.

3.6 HALL EFFECT

If a piece of current-carrying conductor (metal or semiconductor) is placed in a transverse magnetic field, an electric field appears inside the conductor in a direction perpendicular to both the current and the magnetic field. This phenomenon is referred to as the *Hall effect* and the electric field that is created is known as the *Hall field*.

We consider a rectangular specimen of the material carrying a current I in the positive x-direction. Let a magnetic field of induction B be applied along the positive z-direction (Fig. 3.18). A magnetic force will act on the current carriers in the negative y-direction. Consequently the carriers are deflected towards the bottom surface of the specimen and will accumulate there. If the semiconductor is n-type so that the current carriers are electrons, the collection of these electrons on the bottom surface makes it negatively charged relative to the top surface. As a result, an electric field, called the Hall field, appears along the negative y-direction. The force on the current-carrying electrons owing to the Hall field opposes the magnetic force. When the two forces cancel each other, no further accumulation of electrons occurs on the bottom surface. An equilibrium is then established and the Hall field acquires a steady value.

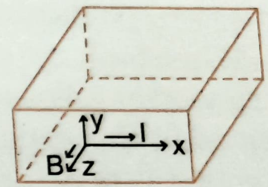

Fig. 3.18 Illustration of the Hall effect.

If the semiconductor is p-type so that the current is carried by holes, the accumulation of the holes on the bottom surface makes this surface positively charged with respect to the top surface.

The Hall field is now developed along the positive y-direction. The force on the holes due to the Hall field is opposite to the magnetic force. In equilibrium, these two forces balance and no further collection of holes is possible. The Hall field then becomes steady.

The average magnetic force on a current carrier is Bev, where e is the charge of a carrier and v is its drift velocity in the x-direction. If F_H is the Hall field developed in the y-direction, the force on a current carrier exerted by this field is eF_H. In equilibrium, the two forces balance, so that

$$e F_H = B ev \tag{3.15}$$

If n_0 is the carrier concentration, the current density in the x-direction is

$$J = n_0 \, ev \tag{3.16}$$

Equations (3.15) and (3.16) give

$$F_H = \frac{J B}{n_0 \, e} \tag{3.17}$$

The quantity $F_H/(JB)$ represents the Hall field per unit current density per unit magnetic field and is known as the *Hall coefficient*, R_H. Thus

$$R_H = \frac{F_H}{J B} = \frac{1}{n_0 \, e}, \tag{3.18}$$

using Eq. (3.17). A rigorous analysis shows that

$$R_H = \frac{r}{n_0 \, e}, \tag{3.19}$$

where r, a constant introduced by the mechanisms scattering the current carriers and the distribution function, is not much different from unity. Thus Eq. (3.18) for R_H does not give much error.

When the current is carried by electrons, the carrier charge is negative so that

$$R_H = -\frac{1}{n e}, \tag{3.20}$$

where n is the electron concentration. Equation (3.20) holds for an n-type semiconductor as well as for a metal.

If the semiconductor is p-type and the current carriers are holes, the carrier charge is positive. Therefore

$$R_H = \frac{1}{p e}, \tag{3.21}$$

where p is the hole concentration. The *sign* of the Hall coefficient is thus *opposite* for n- and p-type semiconductors.

When both the electrons and the holes contribute to the current, the Hall coefficient can be determined as follows. Since the charges of the electrons and the holes are opposite and they are deflected in the same direction by the magnetic field, the net deflection current density in the y-direction is $(e p v_{yp} - e n v_{yn})$, where v_{yp} and v_{yn} denote the y-components of the hole and the electron velocities respectively arising from the deflection by the magnetic field. In equilibrium, this current density is balanced by the current density σF_H produced by the Hall field, σ being the conductivity. In other words,

$$\sigma F_H = e p v_{yp} - e n v_{yn} \tag{3.22}$$

Let v_{xp} and v_{xn} be respectively the drift velocities of the holes and the electrons along the x-direction. The magnetic force on a hole is Bev_{xp} and this produces the velocity v_{yp}. In equilibrium, the magnetic force is balanced by the rate of loss of the y-component of the hole momentum caused by scattering. Thus

$$Bev_{xp} = \frac{m_p \, v_{yp}}{\tau_p} \tag{3.23}$$

where m_p is the effective mass and τ_p is the relaxation time for the holes. Similarly, if m_n and τ_n are respectively the effective mass and the relaxation time for the electrons we have

$$Bev_{xn} = \frac{m_n v_{yn}}{\tau_n} \tag{3.24}$$

If μ_n and μ_p are the electron mobility and the hole mobility, respectively, we can write from Eq. (3.13)

$$\mu_n = \frac{e\tau_n}{m_n} \tag{3.25}$$

and

$$\mu_p = \frac{e\tau_p}{m_p} \tag{3.26}$$

Equations (3.23) and (3.24) then render

$$v_{yp} = \mu_p Bv_{xp} \tag{3.27}$$

and

$$v_{yn} = \mu_n Bv_{xn} \tag{3.28}$$

Substituting for v_{yp} and v_{yn} from Eqs. (3.27) and (3.28) into Eq. (3.22), we get

$$\sigma F_H = e\,p\mu_p Bv_{xp} - e\,n\,\mu_n Bv_{xn} \tag{3.29}$$

$$= \frac{e(p\mu_p^2 - n\mu_n^2)}{\sigma} JB,$$

where v_{xp} and v_{xn} are put equal to $\mu_p F$ and $\mu_n F$, respectively, F being the electric field in the x-direction. Also, the relationship $J = \sigma F$ is used. From Eq. (3.29), we obtain for the Hall coefficient

$$R_H = \frac{F_H}{JB} = \frac{e(p\mu_p^2 - n\mu_n^2)}{\sigma^2} = \frac{p\mu_p^2 - n\mu_n^2}{e(n\mu_n + p\mu_p)^2}, \tag{3.30}$$

since $\sigma = ne\mu_n + pe\mu_p$. When the electrons contribute predominantly, we have $n\mu_n >> p\mu_p$, and Eq. (3.30) gives $R_H = -1/(ne)$, which agrees with Eq. (3.20). When the holes are predominant, we have $p\mu_p >> n\mu_n$, and Eq. (3.30) yields $R_H = 1/(pe)$, agreeing with Eq. (3.21). Note that the Hall coefficient is zero, if $p/n = \mu_n^2/\mu_p^2$.

For an intrinsic semiconductor, $n = p = n_i$, and Eq. (3.30) renders

$$R_H = -\frac{1}{n_i e}\left(\frac{\mu_n - \mu_p}{\mu_n + \mu_p}\right). \tag{3.30A}$$

Experimental determination of the Hall coefficient

A rectangular specimen of the semiconductor of width b and thickness t is placed between the pole pieces of an electromagnet such that a magnetic field B is applied along the z-direction. A current I, measured by the milliammeter A, is sent through the sample in the x-direction by means of a battery (Fig. 3.19). The Hall voltage V_H can be measured with the help of two probes placed at the centres of the top and the bottom surfaces of the sample. We have

$$V_H = F_H t \tag{3.31}$$

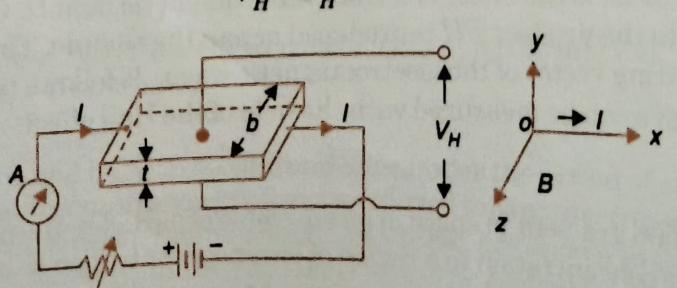

Fig. 3.19 Experimental arrangement for the determination of the Hall coefficient

Since $F_H = R_H JB$, Eq. (3.31) can be written as

$$V_H = R_H JBt \quad (3.32)$$

The current density J is given by

$$J = \frac{I}{tb}, \quad (3.33)$$

the product tb being the cross sectional area of the sample. Substituting for J from Eq. (3.33) into Eq. (3.32), we get

$$V_H = \frac{R_H I B}{b}, \quad \text{or} \quad R_H = \frac{V_H b}{I B} \quad (3.34)$$

Since the quantities on the right-hand side of Eq. (3.34) can be measured, R_H is readily determined. In SI units, V_H is in volt (V), b is in meter (m), I is in ampere (A) and B is in tesla (T). R_H, as obtained from Eq. (3.34), is expressed in m^3/C. Since the polarity of V_H is opposite for n-type and p-type materials, R_H has opposite signs for the two types of semiconductors.

Uses of the Hall effect

(a) *Determination of the semiconductor type*: Since R_H is positive for a p-type semiconductor and negative for an n-type semiconductor, the sign of R_H can ascertain the type of a piece of extrinsic semiconductor.

(b) *Determination of carrier concentration*: From Eqs. (3.20) and (3.21), we obtain $n = 1/|e R_H|$ and $p = 1/(e R_H)$. Thus the electron concentration n and the hole concentration p can be found by measuring R_H.

(c) *Determination of carrier mobility*: If the current is carried by a single type of carriers, say electrons, we have for the conductivity $\sigma = n e\mu_n$. Here, μ_n is the electron mobility. With the aid of Eq. (3.20), we write $\mu_n = \sigma |R_H|$. Thus measurements of σ and R_H give the mobility μ_n. The presence of the factor r in Eq. (3.19) makes the mobility determined from R_H different from the actual *drift mobility*. The mobility given by the product $\sigma |R_H|$ is therefore referred to as the *Hall mobility*. Since the factor r does not differ much from unity, the Hall mobility is nearly equal to the drift mobility.

(d) *Measurement of magnetic field*: The Hall voltage V_H is proportional to the magnetic field B for a given current I through the specimen. Thus knowing the sample dimensions and R_H, the magnetic field can be determined by measuring I and V_H.

(e) *Hall effect multiplier*: If the magnetic field B is produced by a current I' flowing through an air-core coil, B is proportional to I'. The Hall voltage V_H is then proportional to the product II'. Thus the Hall effect serves as the basis for the design of a multiplier.

(f) *Measurement of power in an electromagnetic wave*: The electric field F and the magnetic field H of an electromagnetic wave in free space are perpendicular to each other. If a semiconductor specimen is kept parallel to F, a current I will flow in the semiconductor due to F. At the same time, the semiconductor feels the effect of a transverse magnetic field H. Consequently, a Hall voltage, proportional to the product FH is produced across the sample. The quantity FH is the magnitude of the Poynting vector of the electromagnetic wave. It follows that the power flow in an electromagnetic wave can be measured with the help of the Hall effect.